Secure Micro-Segmentation Report

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1.0 Segmentation Plan and Micro-Segmentation Goals

Planned Security Zones & Segmentation Strategy

Security Zones and Purpose

Finance Zone

- Assets: Finance-VM (192.168.1.2), Finance-VM2 (192.168.1.3)
- Purpose: Stores and processes financial information.

HR Zone

- Assets: HR-VM (192.168.2.2)
- Purpose: Stores employee records and HR-related data.

Database Zone

- Assets: DB-VM (192.168.3.2)
- Purpose: Hosts sensitive databases and application data.

Audit Zone

- Assets: Audit-VM (192.168.4.2)
- Purpose: Provides limited third-party auditing access to Finance logs.

How Segmentation Contributes to Security

Network segmentation divides the network into isolated security zones, reducing the attack surface and limiting lateral movement. If one system becomes compromised, attackers cannot easily access other critical resources. Segmentation also enforces access control policies and improves compliance with security standards.

Micro-Segmentation Goals and Access Controls

The purpose of implementing micro-segmentation is to enforce granular security controls between systems and ensure that only authorized traffic is permitted.

Why Micro-Segmentation is Critical for Zero Trust Security

Micro-segmentation supports the Zero Trust principle of "Never Trust, Always Verify" by restricting communication between systems based on business requirements rather than network location. Every connection must be explicitly authorized.

How Logging Enhances Security Monitoring

Logging records security events and denied traffic attempts. Security teams can use logs to detect unauthorized access attempts, investigate incidents, identify suspicious activity, and improve threat detection capabilities.

2.0 Firewall Rules and Network Traffic Control

Firewall Configuration and Implementation

Firewall rules were configured to allow Audit-VM read-only access to Finance logs while preventing unauthorized communication with other network segments.

Role of Each Rule

- AUDIT-ALLOW permits Audit-VM to access Finance-VM log services.
- AUDIT-DENY blocks Audit-VM from accessing the Database network.
- LOG-DROP records unauthorized traffic attempts and blocks them.

Table of Implemented Rules

Rule Name	Source Subnet	Destination Subnet	Protoc	P	Action

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AUDIT-ALLOW	192.168.4.2	192.168.1.2	UDP	5	Accept
AUDIT-DENY	192.168.4.2	192.168.3.0/24	Any	A	Drop
LOG-DROP	Any	Any	Any	A	Drop + Log

3.0 Traffic Validation and Analysis

Network Connectivity Tests

Connectivity tests were conducted to verify that firewall rules functioned as intended.

Test Results

Allowed Traffic

- Audit-VM successfully accessed Finance-VM logging service on UDP port 514.

Blocked Traffic

- Audit-VM could not access DB-VM.
- Unauthorized traffic was dropped and logged.

Misconfigurations and Resolution

Initially, logging was not enabled for denied traffic. This was corrected by enabling logging on drop rules to provide visibility into unauthorized access attempts.

Capturing and Analyzing Traffic

Traffic was monitored using tcpdump and firewall logs.

Expected Traffic Patterns

- Audit-VM communicating with Finance-VM on UDP 514.
- Normal internal traffic within approved security zones.

Unexpected Traffic Patterns

- Attempts by Audit-VM to access Database resources.
- Any communication outside approved access policies.

Example Captured Data

Source IP	Destination IP	Protocol	P	Allowed/Blocked
192.168.4.2	192.168.1.2	UDP	5	Allowed
192.168.4.2	192.168.3.2	Any	A	Blocked

4.0 Threat Modeling and Risk Analysis

Identifying Vulnerabilities Before Segmentation

Vulnerability 1

Lack of network isolation allowed unrestricted communication between systems.

Vulnerability 2

No centralized logging existed for denied traffic attempts.

Likely Attack Vectors

- Unauthorized lateral movement.
- Privilege escalation.
- Internal reconnaissance and network scanning.
- Data exfiltration from sensitive systems.

Impact on Segmentation Policies

These vulnerabilities increase the risk of attackers moving across network zones and accessing critical data.

Assessing Network Security Risks

Potential Attack Vectors

- Compromised third-party vendor systems.
- Insider threats.
- Credential theft.
- Malware propagation.

Critical Assets Requiring Isolation

- Finance-VM
- Finance-VM2
- DB-VM

Risk Mitigation Through Segmentation

Micro-segmentation restricts communication paths and ensures only approved traffic reaches critical systems.

Applying Threat Modeling to Firewall Policies (STRIDE)

Spoofing

Blocked by allowing only specific source and destination addresses.

Tampering

Restricted access reduces the possibility of unauthorized modifications.

Repudiation

Logging provides evidence of network activity and denied attempts.

Information Disclosure

Access to sensitive systems is restricted using firewall rules.

Denial of Service

Unauthorized traffic is blocked before reaching protected resources.

Elevation of Privilege

Least-privilege access prevents unnecessary permissions.

Firewall Rules Addressing High-Risk Paths

- AUDIT-DENY blocks access to the Database Zone.
- LOG-DROP records all unauthorized traffic attempts.

Limiting Lateral Movement

Segmentation prevents compromised hosts from freely communicating with other zones, significantly reducing attack spread.

Threat Mitigation Adjustments

Recommended Mitigation Strategies

- Maintain default-deny firewall policies.
- Implement multi-factor authentication.

- Regularly review firewall logs.
- Conduct periodic vulnerability assessments.

Logging and Monitoring Improvements

- Enable real-time alerts for repeated denied traffic.
- Forward logs to a SIEM platform.
- Create dashboards for security monitoring.

Additional Security Controls

- Intrusion Detection Systems (IDS).
- Intrusion Prevention Systems (IPS).
- Network Access Control (NAC).
- Security Information and Event Management (SIEM).

5.0 Reflection on Security Implementation

Effectiveness of Micro-Segmentation

The implemented rules successfully enforced segmentation and logging requirements. Authorized access was permitted while unauthorized traffic was blocked and logged.

Security Improvements

- Reduced attack surface.
- Enhanced visibility through logging.
- Improved compliance with Zero Trust principles.
- Better protection of critical assets.

Challenges and Adjustments

Challenges Encountered

- Configuring correct firewall rule order.
- Ensuring logging did not affect network performance.
- Testing communication between multiple zones.

Troubleshooting Process

- Reviewed firewall configurations.
- Verified IP addressing and routing.
- Examined firewall logs and tcpdump captures.

Future Security Enhancements

Recommended Improvements

- Implement SIEM integration.
- Automate security monitoring and alerting.
- Deploy IDS/IPS technologies.
- Apply role-based access control.

Automated Monitoring

Security monitoring can be automated through SIEM platforms that collect, analyze, and generate alerts based on suspicious activities.

6.0 Broader Security Implications

Micro-segmentation is a key enterprise security strategy because it minimizes attack surfaces and prevents attackers from moving laterally across networks. Controlled access strengthens Zero Trust security by ensuring every communication request is verified and authorized.

Real-world threats mitigated by micro-segmentation include:

- Ransomware propagation.
- Insider threats.
- Credential theft attacks.
- Unauthorized third-party access.
- Data exfiltration attempts.
- Advanced Persistent Threats (APTs).

By implementing micro-segmentation, SecureCorp significantly improved network visibility, access control, and overall security posture while aligning with Zero Trust security principles.